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Autonomous drug delivery system

Abstract

An autonomous drug delivery system advantageously utilizes physiological monitor outputs so as to automatically give a bolus of a rescue drug or other necessary medication when certain criteria and confidence levels are met. An emergency button is provided to manually trigger administration of the rescue drug. The rescue drug may be an opioid antagonist in response to an analgesia overdose, a hypotensive drug to avert an excessive drop in blood pressure or an anti-arrhythmia drug to suppress abnormal heartbeats, to name a few.

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7873497	12/2010	Weber et al.	N/A	N/A
7880606	12/2010	Al-Ali	N/A	N/A
7880626	12/2010	Al-Ali et al.	N/A	N/A

7891355	12/2010	Al-Ali et al.	N/A	N/A
7894868	12/2010	Al-Ali et al.	N/A	N/A
7899507	12/2010	Al-Ali et al.	N/A	N/A
7904132	12/2010	Weber et al.	N/A	N/A
7909772	12/2010	Popov et al.	N/A	N/A
7910875	12/2010	Al-Ali	N/A	N/A
7919713	12/2010	Al-Ali et al.	N/A	N/A
7937128	12/2010	Al-Ali	N/A	N/A
7937129	12/2010	Mason et al.	N/A	N/A
7937130	12/2010	Diab et al.	N/A	N/A
7941199	12/2010	Kiani	N/A	N/A
7951086	12/2010	Flaherty et al.	N/A	N/A
7957780	12/2010	Lamego et al.	N/A	N/A
7962188	12/2010	Kiani et al.	N/A	N/A
7962190	12/2010	Diab et al.	N/A	N/A
7976472	12/2010	Kiani	N/A	N/A
7988637	12/2010	Diab	N/A	N/A
7990382	12/2010	Kiani	N/A	N/A
7991446	12/2010	Ali et al.	N/A	N/A
8000761	12/2010	Al-Ali	N/A	N/A
8008088	12/2010	Bellott et al.	N/A	N/A
RE42753	12/2010	Kiani- Azarbayjany et al.	N/A	N/A
8019400	12/2010	Diab et al.	N/A	N/A
8028701	12/2010	Al-Ali et al.	N/A	N/A
8029765	12/2010	Bellott et al.	N/A	N/A
8036727	12/2010	Schurman et al.	N/A	N/A
8036728	12/2010	Diab et al.	N/A	N/A
8046040	12/2010	Ali et al.	N/A	N/A
8046041	12/2010	Diab et al.	N/A	N/A
8046042	12/2010	Diab et al.	N/A	N/A
8048040	12/2010	Kiani	N/A	N/A
8050728	12/2010	Al-Ali et al.	N/A	N/A
RE43169	12/2011	Parker	N/A	N/A
8118620	12/2011	Al-Ali et al.	N/A	N/A
8126528	12/2011	Diab et al.	N/A	N/A
8128572	12/2011	Diab et al.	N/A	N/A
8130105	12/2011	Al-Ali et al.	N/A	N/A
8145287	12/2011	Diab et al.	N/A	N/A
8150487	12/2011	Diab et al.	N/A	N/A
8175672	12/2011	Parker	N/A	N/A
8180420	12/2011	Diab et al.	N/A	N/A
8182443	12/2011	Kiani	N/A	N/A
8185180	12/2011	Diab et al.	N/A	N/A
8190223	12/2011	Al-Ali et al.	N/A	N/A
8190227	12/2011	Diab et al.	N/A	N/A
8203438	12/2011	Kiani et al.	N/A	N/A
8203704	12/2011	Merritt et al.	N/A	N/A
8204566	12/2011	Schurman et al.	N/A	N/A

8219172	12/2011	Schurman et al.	N/A	N/A
8224411	12/2011	Al-Ali et al.	N/A	N/A
8228181	12/2011	Ai-Ali	N/A	N/A
8229532	12/2011	Davis	N/A	N/A
8229533	12/2011	Diab et al.	N/A	N/A
8233955	12/2011	Al-Ali et al.	N/A	N/A
8244325	12/2011	Al-Ali et al.	N/A	N/A
8255026	12/2011	Al-Ali	N/A	N/A
8255027	12/2011	Al-Ali et al.	N/A	N/A
8255028	12/2011	Al-Ali et al.	N/A	N/A
8260577	12/2011	Weber et al.	N/A	N/A
8265723	12/2011	McHale et al.	N/A	N/A
8274360	12/2011	Sampath et al.	N/A	N/A
8280473	12/2011	Al-Ali	N/A	N/A
8301217	12/2011	Al-Ali et al.	N/A	N/A
8306596	12/2011	Schurman et al.	N/A	N/A
8310336	12/2011	Muhsin et al.	N/A	N/A
8315683	12/2011	Al-Ali et al.	N/A	N/A
RE43860	12/2011	Parker	N/A	N/A
8337403	12/2011	Al-Ali et al.	N/A	N/A
8346330	12/2012	Lamego	N/A	N/A
8353842	12/2012	Al-Ali et al.	N/A	N/A
8355766	12/2012	MacNeish, III et al.	N/A	N/A
8359080	12/2012	Diab et al.	N/A	N/A
8364223	12/2012	Al-Ali et al.	N/A	N/A
8364226	12/2012	Diab et al.	N/A	N/A
8374665	12/2012	Lamego	N/A	N/A
8385995	12/2012	Al-Ali et al.	N/A	N/A
8385996	12/2012	Smith et al.	N/A	N/A
8388353	12/2012	Kiani et al.	N/A	N/A
8399822	12/2012	Al-Ali	N/A	N/A
8401602	12/2012	Kiani	N/A	N/A
8405608	12/2012	Al-Ali et al.	N/A	N/A
8414499	12/2012	Al-Ali et al.	N/A	N/A
8418524	12/2012	Al-Ali	N/A	N/A
8423106	12/2012	Lamego et al.	N/A	N/A
8428967	12/2012	Olsen et al.	N/A	N/A
8430817	12/2012	Al-Ali et al.	N/A	N/A
8437825	12/2012	Dalvi et al.	N/A	N/A
8455290	12/2012	Siskavich	N/A	N/A
8457703	12/2012	Al-Ali	N/A	N/A
8457707	12/2012	Kiani	N/A	N/A
8463349	12/2012	Diab et al.	N/A	N/A
8466286	12/2012	Bellott et al.	N/A	N/A
8471713	12/2012	Poeze et al.	N/A	N/A
8473020	12/2012	Kiani et al.	N/A	N/A
8483787	12/2012	Al-Ali et al.	N/A	N/A
8489364	12/2012	Weber et al.	N/A	N/A
8498684	12/2012	Weber et al.	N/A	N/A

8504128	12/2012	Blank et al.	N/A	N/A
8509867	12/2012	Workman et al.	N/A	N/A
8515509	12/2012	Bruinsma et al.	N/A	N/A
8523781	12/2012	Al-Ali	N/A	N/A
8529301	12/2012	Al-Ali et al.	N/A	N/A
8532727	12/2012	Ali et al.	N/A	N/A
8532728	12/2012	Diab et al.	N/A	N/A
D692145	12/2012	Al-Ali et al.	N/A	N/A
8547209	12/2012	Kiani et al.	N/A	N/A
8548548	12/2012	Al-Ali	N/A	N/A
8548549	12/2012	Schurman et al.	N/A	N/A
8548550	12/2012	Al-Ali et al.	N/A	N/A
8560032	12/2012	Al-Ali et al.	N/A	N/A
8560034	12/2012	Diab et al.	N/A	N/A
8570167	12/2012	Al-Ali	N/A	N/A
8570503	12/2012	Vo et al.	N/A	N/A
8571617	12/2012	Reichgott et al.	N/A	N/A
8571618	12/2012	Lamego et al.	N/A	N/A
8571619	12/2012	Al-Ali et al.	N/A	N/A
8577431	12/2012	Lamego et al.	N/A	N/A
8581732	12/2012	Al-Ali et al.	N/A	N/A
8584345	12/2012	Al-Ali et al.	N/A	N/A
8588880	12/2012	Abdul-Hafiz et al.	N/A	N/A
8600467	12/2012	Al-Ali et al.	N/A	N/A
8606342	12/2012	Diab	N/A	N/A
8626255	12/2013	Al-Ali et al.	N/A	N/A
8630691	12/2013	Lamego et al.	N/A	N/A
8634889	12/2013	Al-Ali et al.	N/A	N/A
8641631	12/2013	Sierra et al.	N/A	N/A
8652060	12/2013	Al-Ali	N/A	N/A
8663107	12/2013	Kiani	N/A	N/A
8666468	12/2013	Al-Ali	N/A	N/A
8667967	12/2013	Al-Ali et al.	N/A	N/A
8670811	12/2013	O'Reilly	N/A	N/A
8670814	12/2013	Diab et al.	N/A	N/A
8676286	12/2013	Weber et al.	N/A	N/A
8682407	12/2013	Al-Ali	N/A	N/A
RE44823	12/2013	Parker	N/A	N/A
RE44875	12/2013	Kiani et al.	N/A	N/A
8688183	12/2013	Bruinsma et al.	N/A	N/A
8690799	12/2013	Telfort et al.	N/A	N/A
8700112	12/2013	Kiani	N/A	N/A
8702627	12/2013	Telfort et al.	N/A	N/A
8706179	12/2013	Parker	N/A	N/A
8712494	12/2013	MacNeish, III et al.	N/A	N/A
8715206	12/2013	Telfort et al.	N/A	N/A
8718735	12/2013	Lamego et al.	N/A	N/A
8718737	12/2013	Diab et al.	N/A	N/A
8718738	12/2013	Blank et al.	N/A	N/A

8720249	12/2013	Al-Ali	N/A	N/A
8721541	12/2013	Al-Ali et al.	N/A	N/A
8721542	12/2013	Al-Ali et al.	N/A	N/A
8723677	12/2013	Kiani	N/A	N/A
8740792	12/2013	Kiani et al.	N/A	N/A
8754776	12/2013	Poeze et al.	N/A	N/A
8755535	12/2013	Telfort et al.	N/A	N/A
8755856	12/2013	Diab et al.	N/A	N/A
8755872	12/2013	Marinow	N/A	N/A
8761850	12/2013	Lamego	N/A	N/A
8764671	12/2013	Kiani	N/A	N/A
8768423	12/2013	Shakespeare et al.	N/A	N/A
8771204	12/2013	Telfort et al.	N/A	N/A
8777634	12/2013	Kiani et al.	N/A	N/A
8781543	12/2013	Diab et al.	N/A	N/A
8781544	12/2013	Al-Ali et al.	N/A	N/A
8781549	12/2013	Al-Ali et al.	N/A	N/A
8788003	12/2013	Schurman et al.	N/A	N/A
8790268	12/2013	Al-Ali	N/A	N/A
8801613	12/2013	Al-Ali et al.	N/A	N/A
8821397	12/2013	Al-Ali et al.	N/A	N/A
8821415	12/2013	Al-Ali et al.	N/A	N/A
8830449	12/2013	Lamego et al.	N/A	N/A
8831700	12/2013	Schurman et al.	N/A	N/A
8840549	12/2013	Al-Ali et al.	N/A	N/A
8847740	12/2013	Kiani et al.	N/A	N/A
8849365	12/2013	Smith et al.	N/A	N/A
8852094	12/2013	Al-Ali et al.	N/A	N/A
8852994	12/2013	Wojtczuk et al.	N/A	N/A
8868147	12/2013	Stippick et al.	N/A	N/A
8868150	12/2013	Al-Ali et al.	N/A	N/A
8870792	12/2013	Al-Ali et al.	N/A	N/A
8886271	12/2013	Kiani et al.	N/A	N/A
8888539	12/2013	Al-Ali et al.	N/A	N/A
8888708	12/2013	Diab et al.	N/A	N/A
8892180	12/2013	Weber et al.	N/A	N/A
8897847	12/2013	Al-Ali	N/A	N/A
8909310	12/2013	Lamego et al.	N/A	N/A
8911377	12/2013	Al-Ali	N/A	N/A
8912909	12/2013	Al-Ali et al.	N/A	N/A
8920317	12/2013	Al-Ali et al.	N/A	N/A
8921699	12/2013	Al-Ali et al.	N/A	N/A
8922382	12/2013	Al-Ali et al.	N/A	N/A
8929964	12/2014	Al-Ali et al.	N/A	N/A
8942777	12/2014	Diab et al.	N/A	N/A
8948834	12/2014	Diab et al.	N/A	N/A
8948835	12/2014	Diab	N/A	N/A
8965471	12/2014	Lamego	N/A	N/A
8983564	12/2014	Al-Ali	N/A	N/A
8989831	12/2014	Al-Ali et al.	N/A	N/A

8996085	12/2014	Kiani et al.	N/A	N/A
8998809	12/2014	Kiani	N/A	N/A
9028429	12/2014	Telfort et al.	N/A	N/A
9037207	12/2014	Al-Ali et al.	N/A	N/A
9060721	12/2014	Reichgott et al.	N/A	N/A
9066666	12/2014	Kiani	N/A	N/A
9066680	12/2014	Al-Ali et al.	N/A	N/A
9072474	12/2014	Al-Ali et al.	N/A	N/A
9078560	12/2014	Schurman et al.	N/A	N/A
9084569	12/2014	Weber et al.	N/A	N/A
9095316	12/2014	Welch et al.	N/A	N/A
9106038	12/2014	Telfort et al.	N/A	N/A
9107625	12/2014	Telfort et al.	N/A	N/A
9107626	12/2014	Al-Ali et al.	N/A	N/A
9113831	12/2014	Al-Ali	N/A	N/A
9113832	12/2014	Al-Ali	N/A	N/A
9119595	12/2014	Lamego	N/A	N/A
9131881	12/2014	Diab et al.	N/A	N/A
9131882	12/2014	Al-Ali et al.	N/A	N/A
9131883	12/2014	Al-Ali	N/A	N/A
9131917	12/2014	Telfort et al.	N/A	N/A
9138180	12/2014	Coverston et al.	N/A	N/A
9138182	12/2014	Al-Ali et al.	N/A	N/A
9138192	12/2014	Weber et al.	N/A	N/A
9142117	12/2014	Muhsin et al.	N/A	N/A
9153112	12/2014	Kiani et al.	N/A	N/A
9153121	12/2014	Kiani et al.	N/A	N/A
9161696	12/2014	Al-Ali et al.	N/A	N/A
9161713	12/2014	Al-Ali et al.	N/A	N/A
9167995	12/2014	Lamego et al.	N/A	N/A
9176141	12/2014	Al-Ali et al.	N/A	N/A
9186102	12/2014	Bruinsma et al.	N/A	N/A
9192312	12/2014	Al-Ali	N/A	N/A
9192329	12/2014	Al-Ali	N/A	N/A
9192351	12/2014	Telfort et al.	N/A	N/A
9195385	12/2014	Al-Ali et al.	N/A	N/A
9211072	12/2014	Kiani	N/A	N/A
9211095	12/2014	Al-Ali	N/A	N/A
9218454	12/2014	Kiani et al.	N/A	N/A
9226696	12/2015	Kiani	N/A	N/A
9241662	12/2015	Al-Ali et al.	N/A	N/A
9245668	12/2015	Vo et al.	N/A	N/A
9259185	12/2015	Abdul-Hafiz et al.	N/A	N/A
9267572	12/2015	Barker et al.	N/A	N/A
9277880	12/2015	Poeze et al.	N/A	N/A
9289167	12/2015	Diab et al.	N/A	N/A
9295421	12/2015	Kiani et al.	N/A	N/A
9307928	12/2015	Al-Ali et al.	N/A	N/A
9323894	12/2015	Kiani	N/A	N/A
D755392	12/2015	Hwang et al.	N/A	N/A

9326712	12/2015	Kiani	N/A	N/A
9333316	12/2015	Kiani	N/A	N/A
9339220	12/2015	Lamego et al.	N/A	N/A
9341565	12/2015	Lamego et al.	N/A	N/A
9351673	12/2015	Diab et al.	N/A	N/A
9351675	12/2015	Al-Ali et al.	N/A	N/A
9364181	12/2015	Kiani et al.	N/A	N/A
9368671	12/2015	Wojtczuk et al.	N/A	N/A
9370325	12/2015	Al-Ali et al.	N/A	N/A
9370326	12/2015	McHale et al.	N/A	N/A
9370335	12/2015	Al-Ali et al.	N/A	N/A
9375185	12/2015	Ali et al.	N/A	N/A
9386953	12/2015	Al-Ali	N/A	N/A
9386961	12/2015	Al-Ali et al.	N/A	N/A
9392945	12/2015	Al-Ali et al.	N/A	N/A
9397448	12/2015	Al-Ali et al.	N/A	N/A
9408542	12/2015	Kinast et al.	N/A	N/A
9436645	12/2015	Al-Ali et al.	N/A	N/A
9445759	12/2015	Lamego et al.	N/A	N/A
9466919	12/2015	Kiani et al.	N/A	N/A
9474474	12/2015	Lamego et al.	N/A	N/A
9480422	12/2015	Al-Ali	N/A	N/A
9480435	12/2015	Olsen	N/A	N/A
9492110	12/2015	Al-Ali et al.	N/A	N/A
9510779	12/2015	Poeze et al.	N/A	N/A
9517024	12/2015	Kiani et al.	N/A	N/A
9532722	12/2016	Lamego et al.	N/A	N/A
9538949	12/2016	Al-Ali et al.	N/A	N/A
9538980	12/2016	Telfort et al.	N/A	N/A
9549696	12/2016	Lamego et al.	N/A	N/A
9554737	12/2016	Schurman et al.	N/A	N/A
9560996	12/2016	Kiani	N/A	N/A
9560998	12/2016	Al-Ali et al.	N/A	N/A
9566019	12/2016	Al-Ali et al.	N/A	N/A
9579039	12/2016	Jansen et al.	N/A	N/A
9591975	12/2016	Dalvi et al.	N/A	N/A
9622692	12/2016	Lamego et al.	N/A	N/A
9622693	12/2016	Diab	N/A	N/A
D788312	12/2016	Al-Ali et al.	N/A	N/A
9649054	12/2016	Lamego et al.	N/A	N/A
9697928	12/2016	Al-Ali et al.	N/A	N/A
9717458	12/2016	Lamego et al.	N/A	N/A
9724016	12/2016	Al-Ali et al.	N/A	N/A
9724024	12/2016	Al-Ali	N/A	N/A
9724025	12/2016	Kiani et al.	N/A	N/A
9749232	12/2016	Sampath et al.	N/A	N/A
9750442	12/2016	Olsen	N/A	N/A
9750461	12/2016	Telfort	N/A	N/A
9775545	12/2016	Al-Ali et al.	N/A	N/A
9778079	12/2016	Al-Ali et al.	N/A	N/A

9782077	12/2016	Lamego et al.	N/A	N/A
9787568	12/2016	Lamego et al.	N/A	N/A
9808188	12/2016	Perea et al.	N/A	N/A
9839379	12/2016	Al-Ali et al.	N/A	N/A
9839381	12/2016	Weber et al.	N/A	N/A
9847749	12/2016	Kiani et al.	N/A	N/A
9848800	12/2016	Lee et al.	N/A	N/A
9861298	12/2017	Eckerbom et al.	N/A	N/A
9861305	12/2017	Weber et al.	N/A	N/A
9877650	12/2017	Muhsin et al.	N/A	N/A
9891079	12/2017	Dalvi	N/A	N/A
9924897	12/2017	Abdul-Hafiz	N/A	N/A
9936917	12/2017	Poeze et al.	N/A	N/A
9955937	12/2017	Telfort	N/A	N/A
9965946	12/2017	Al-Ali et al.	N/A	N/A
D820865	12/2017	Muhsin et al.	N/A	N/A
9986952	12/2017	Dalvi et al.	N/A	N/A
D822215	12/2017	Al-Ali et al.	N/A	N/A
D822216	12/2017	Barker et al.	N/A	N/A
10010276	12/2017	Al-Ali et al.	N/A	N/A
10086138	12/2017	Novak, Jr.	N/A	N/A
10111591	12/2017	Dyell et al.	N/A	N/A
D833624	12/2017	DeJong et al.	N/A	N/A
10123729	12/2017	Dyell et al.	N/A	N/A
D835282	12/2017	Barker et al.	N/A	N/A
D835283	12/2017	Barker et al.	N/A	N/A
D835284	12/2017	Barker et al.	N/A	N/A
D835285	12/2017	Barker et al.	N/A	N/A
10149616	12/2017	Al-Ali et al.	N/A	N/A
10154815	12/2017	Al-Ali et al.	N/A	N/A
10159412	12/2017	Lamego et al.	N/A	N/A
10188348	12/2018	Al-Ali et al.	N/A	N/A
RE47218	12/2018	Al-Ali	N/A	N/A
RE47244	12/2018	Kiani et al.	N/A	N/A
RE47249	12/2018	Kiani et al.	N/A	N/A
10205291	12/2018	Scruggs et al.	N/A	N/A
10226187	12/2018	Al-Ali et al.	N/A	N/A
10231657	12/2018	Al-Ali et al.	N/A	N/A
10231670	12/2018	Blank et al.	N/A	N/A
RE47353	12/2018	Kiani et al.	N/A	N/A
10279247	12/2018	Kiani	N/A	N/A
10292664	12/2018	Al-Ali	N/A	N/A
10299720	12/2018	Brown et al.	N/A	N/A
10327337	12/2018	Schmidt et al.	N/A	N/A
10327713	12/2018	Barker et al.	N/A	N/A
10332630	12/2018	Al-Ali	N/A	N/A
10383520	12/2018	Wojtczuk et al.	N/A	N/A
10383527	12/2018	Al-Ali	N/A	N/A
10388120	12/2018	Muhsin et al.	N/A	N/A
D864120	12/2018	Forrest et al.	N/A	N/A

10441181	12/2018	Telfort et al.	N/A	N/A
10441196	12/2018	Eckerbom et al.	N/A	N/A
10448844	12/2018	Al-Ali et al.	N/A	N/A
10448871	12/2018	Al-Ali et al.	N/A	N/A
10456038	12/2018	Lamego et al.	N/A	N/A
10463340	12/2018	Telfort et al.	N/A	N/A
10471159	12/2018	Lapotko et al.	N/A	N/A
10505311	12/2018	Al-Ali et al.	N/A	N/A
10524738	12/2019	Olsen	N/A	N/A
10532174	12/2019	Al-Ali	N/A	N/A
10537285	12/2019	Shreim et al.	N/A	N/A
10542903	12/2019	Al-Ali et al.	N/A	N/A
10555678	12/2019	Dalvi et al.	N/A	N/A
10568553	12/2019	O'Neil et al.	N/A	N/A
RE47882	12/2019	Al-Ali	N/A	N/A
10608817	12/2019	Haider et al.	N/A	N/A
D880477	12/2019	Forrest et al.	N/A	N/A
10617302	12/2019	Al-Ali et al.	N/A	N/A
10617335	12/2019	Al-Ali et al.	N/A	N/A
10637181	12/2019	Al-Ali et al.	N/A	N/A
D886849	12/2019	Muhsin et al.	N/A	N/A
D887548	12/2019	Abdul-Hafiz et al.	N/A	N/A
D887549	12/2019	Abdul-Hafiz et al.	N/A	N/A
10667764	12/2019	Ahmed et al.	N/A	N/A
D890708	12/2019	Forrest et al.	N/A	N/A
10721785	12/2019	Al-Ali	N/A	N/A
10736518	12/2019	Al-Ali et al.	N/A	N/A
10750984	12/2019	Pauley et al.	N/A	N/A
D897098	12/2019	Al-Ali	N/A	N/A
10779098	12/2019	Iswanto et al.	N/A	N/A
10827961	12/2019	Iyengar et al.	N/A	N/A
10828007	12/2019	Telfort et al.	N/A	N/A
10832818	12/2019	Muhsin et al.	N/A	N/A
10849554	12/2019	Shreim et al.	N/A	N/A
10856750	12/2019	Indorf	N/A	N/A
D906970	12/2020	Forrest et al.	N/A	N/A
D908213	12/2020	Abdul-Hafiz et al.	N/A	N/A
10918281	12/2020	Al-Ali et al.	N/A	N/A
10932705	12/2020	Muhsin et al.	N/A	N/A
10932729	12/2020	Kiani et al.	N/A	N/A
10939878	12/2020	Kiani et al.	N/A	N/A
10956950	12/2020	Al-Ali et al.	N/A	N/A
D916135	12/2020	Indorf et al.	N/A	N/A
D917046	12/2020	Abdul-Hafiz et al.	N/A	N/A
D917550	12/2020	Indorf et al.	N/A	N/A
D917564	12/2020	Indorf et al.	N/A	N/A
D917704	12/2020	Al-Ali et al.	N/A	N/A
10987066	12/2020	Chandran et al.	N/A	N/A
10991135	12/2020	Al-Ali et al.	N/A	N/A
D919094	12/2020	Al-Ali et al.	N/A	N/A

D919100	12/2020	Al-Ali et al.	N/A	N/A
11006867	12/2020	Al-Ali	N/A	N/A
D921202	12/2020	Al-Ali et al.	N/A	N/A
11024064	12/2020	Muhsin et al.	N/A	N/A
11026604	12/2020	Chen et al.	N/A	N/A
D925597	12/2020	Chandran et al.	N/A	N/A
D927699	12/2020	Al-Ali et al.	N/A	N/A
11076777	12/2020	Lee et al.	N/A	N/A
11114188	12/2020	Poeze et al.	N/A	N/A
D933232	12/2020	Al-Ali et al.	N/A	N/A
D933233	12/2020	Al-Ali et al.	N/A	N/A
D933234	12/2020	Al-Ali et al.	N/A	N/A
11145408	12/2020	Sampath et al.	N/A	N/A
11147518	12/2020	Al-Ali et al.	N/A	N/A
11185262	12/2020	Al-Ali et al.	N/A	N/A
11191484	12/2020	Kiani et al.	N/A	N/A
D946596	12/2021	Ahmed	N/A	N/A
D946597	12/2021	Ahmed	N/A	N/A
D946598	12/2021	Ahmed	N/A	N/A
D946617	12/2021	Ahmed	N/A	N/A
11259745	12/2021	Novak, Jr.	N/A	N/A
11272839	12/2021	Al-Ali et al.	N/A	N/A
11289199	12/2021	Al-Ali	N/A	N/A
RE49034	12/2021	Al-Ali	N/A	N/A
11298021	12/2021	Muhsin et al.	N/A	N/A
D950580	12/2021	Ahmed	N/A	N/A
D950599	12/2021	Ahmed	N/A	N/A
D950738	12/2021	Al-Ali et al.	N/A	N/A
D957648	12/2021	Al-Ali	N/A	N/A
11382567	12/2021	O'Brien et al.	N/A	N/A
11389093	12/2021	Triman et al.	N/A	N/A
11406286	12/2021	Al-Ali et al.	N/A	N/A
11417426	12/2021	Muhsin et al.	N/A	N/A
11439329	12/2021	Lamego	N/A	N/A
11445948	12/2021	Scruggs et al.	N/A	N/A
D965789	12/2021	Al-Ali et al.	N/A	N/A
D967433	12/2021	Al-Ali et al.	N/A	N/A
11464410	12/2021	Muhsin	N/A	N/A
11504058	12/2021	Sharma et al.	N/A	N/A
11504066	12/2021	Dalvi et al.	N/A	N/A
D971933	12/2021	Ahmed	N/A	N/A
D973072	12/2021	Ahmed	N/A	N/A
D973685	12/2021	Ahmed	N/A	N/A
D973686	12/2021	Ahmed	N/A	N/A
D974193	12/2022	Forrest et al.	N/A	N/A
D979516	12/2022	Al-Ali et al.	N/A	N/A
D980091	12/2022	Forrest et al.	N/A	N/A
11596363	12/2022	Lamego	N/A	N/A
11627919	12/2022	Kiani et al.	N/A	N/A
11637437	12/2022	Al-Ali et al.	N/A	N/A

D985498	12/2022	Al-Ali et al.	N/A	N/A
11653862	12/2022	Dalvi et al.	N/A	N/A
D989112	12/2022	Muhsin et al.	N/A	N/A
D989327	12/2022	Al-Ali et al.	N/A	N/A
11678829	12/2022	Al-Ali et al.	N/A	N/A
11679579	12/2022	Al-Ali	N/A	N/A
11684296	12/2022	Vo et al.	N/A	N/A
11692934	12/2022	Normand et al.	N/A	N/A
11701043	12/2022	Al-Ali et al.	N/A	N/A
D997365	12/2022	Hwang	N/A	N/A
11721105	12/2022	Ranasinghe et al.	N/A	N/A
11730379	12/2022	Ahmed et al.	N/A	N/A
D998625	12/2022	Indorf et al.	N/A	N/A
D998630	12/2022	Indorf et al.	N/A	N/A
D998631	12/2022	Indorf et al.	N/A	N/A
D999244	12/2022	Indorf et al.	N/A	N/A
D999245	12/2022	Indorf et al.	N/A	N/A
D999246	12/2022	Indorf et al.	N/A	N/A
11766198	12/2022	Pauley et al.	N/A	N/A
D1000975	12/2022	Al-Ali et al.	N/A	N/A
11803623	12/2022	Kiani et al.	N/A	N/A
11832940	12/2022	Diab et al.	N/A	N/A
D1013179	12/2023	Al-Ali et al.	N/A	N/A
11872156	12/2023	Telfort et al.	N/A	N/A
11879960	12/2023	Ranasinghe et al.	N/A	N/A
11883129	12/2023	Olsen	N/A	N/A
11883190	12/2023	Novak, Jr.	N/A	N/A
2001/0034477	12/2000	Mansfield et al.	N/A	N/A
2001/0039483	12/2000	Brand et al.	N/A	N/A
2002/0010401	12/2001	Bushmakina et al.	N/A	N/A
2002/0058864	12/2001	Mansfield et al.	N/A	N/A
2002/0133080	12/2001	Apruzzese et al.	N/A	N/A
2002/0151816	12/2001	Rich	600/547	A61B 5/02028
2003/0013975	12/2002	Kiani	N/A	N/A
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2004/0117204	12/2003	Mazar	705/2	G16H 40/63
2005/0055276	12/2004	Kiani et al.	N/A	N/A
2005/0182389	12/2004	LaPorte et al.	N/A	N/A
2005/0234317	12/2004	Kiani	N/A	N/A
2006/0073719	12/2005	Kiani	N/A	N/A
2006/0189871	12/2005	Al-Ali et al.	N/A	N/A
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2007/0180140	12/2006	Welch et al.	N/A	N/A
2007/0244377	12/2006	Cozad et al.	N/A	N/A
2007/0282478	12/2006	Al-Ali et al.	N/A	N/A
2008/0064965	12/2007	Jay et al.	N/A	N/A
2008/0094228	12/2007	Welch et al.	N/A	N/A
2008/0103375	12/2007	Kiani	N/A	N/A
2008/0221418	12/2007	Al-Ali et al.	N/A	N/A
2008/0300572	12/2007	Rankers et al.	N/A	N/A
2009/0036759	12/2008	Ault et al.	N/A	N/A
2009/0062730	12/2008	Woo	N/A	N/A
2009/0069787	12/2008	Estes et al.	N/A	N/A
2009/0093687	12/2008	Telfort et al.	N/A	N/A
2009/0095926	12/2008	MacNeish, III	N/A	N/A
2009/0099427	12/2008	Jina et al.	N/A	N/A
2009/0149839	12/2008	Hyde et al.	N/A	N/A
2009/0247984	12/2008	Lamego et al.	N/A	N/A
2009/0259217	12/2008	Hyde et al.	N/A	N/A
2009/0275813	12/2008	Davis	N/A	N/A
2009/0275844	12/2008	Al-Ali	N/A	N/A
2010/0004518	12/2009	Vo et al.	N/A	N/A
2010/0030040	12/2009	Poeze et al.	N/A	N/A
2010/0081942	12/2009	Huiku	N/A	N/A
2010/0099964	12/2009	O'Reilly et al.	N/A	N/A
2010/0185143	12/2009	Uhland et al.	N/A	N/A
2010/0234718	12/2009	Sampath et al.	N/A	N/A
2010/0270257	12/2009	Wachman et al.	N/A	N/A
2011/0028806	12/2010	Merritt et al.	N/A	N/A
2011/0028809	12/2010	Goodman	N/A	N/A
2011/0040197	12/2010	Welch et al.	N/A	N/A
2011/0071464	12/2010	Palerm	604/66	A61M 5/1723
2011/0082711	12/2010	Poeze et al.	N/A	N/A
2011/0087081	12/2010	Kiani et al.	N/A	N/A
2011/0105854	12/2010	Kiani et al.	N/A	N/A
2011/0118561	12/2010	Tari et al.	N/A	N/A
2011/0125060	12/2010	Telfort et al.	N/A	N/A
2011/0137297	12/2010	Kiani et al.	N/A	N/A
2011/0152651	12/2010	Berkow	N/A	N/A
2011/0172498	12/2010	Olsen et al.	N/A	N/A
2011/0208015	12/2010	Welch et al.	N/A	N/A
2011/0213212	12/2010	Al-Ali	N/A	N/A
2011/0230733	12/2010	Al-Ali	N/A	N/A
2011/0237969	12/2010	Eckerbom et al.	N/A	N/A
2011/0288383	12/2010	Diab	N/A	N/A
2012/0022443	12/2011	Robertson et al.	N/A	N/A
2012/0041316	12/2011	Al Ali et al.	N/A	N/A
2012/0046557	12/2011	Kiani	N/A	N/A
2012/0059267	12/2011	Lamego et al.	N/A	N/A
2012/0088984	12/2011	Al-Ali et al.	N/A	N/A

2012/0123231	12/2011	O'Reilly	N/A	N/A
2012/0165629	12/2011	Merritt et al.	N/A	N/A
2012/0185267	12/2011	Kamen et al.	N/A	N/A
2012/0209082	12/2011	Al-Ali	N/A	N/A
2012/0209084	12/2011	Olsen et al.	N/A	N/A
2012/0226117	12/2011	Lamego et al.	N/A	N/A
2012/0259282	12/2011	Alderete, Jr. et al.	N/A	N/A
2012/0283524	12/2011	Kiani et al.	N/A	N/A
2012/0296178	12/2011	Lamego et al.	N/A	N/A
2012/0319816	12/2011	Al-Ali	N/A	N/A
2013/0023775	12/2012	Lamego et al.	N/A	N/A
2013/0041591	12/2012	Lamego	N/A	N/A
2013/0046204	12/2012	Lamego et al.	N/A	N/A
2013/0060147	12/2012	Welch et al.	N/A	N/A
2013/0096405	12/2012	Garfio	N/A	N/A
2013/0096936	12/2012	Sampath et al.	N/A	N/A
2013/0243021	12/2012	Siskavich	N/A	N/A
2013/0253334	12/2012	Al-Ali et al.	N/A	N/A
2013/0267804	12/2012	Al-Ali	N/A	N/A
2013/0274572	12/2012	Al-Ali et al.	N/A	N/A
2013/0296672	12/2012	O'Neil et al.	N/A	N/A
2013/0296713	12/2012	Al-Ali et al.	N/A	N/A
2013/0324808	12/2012	Al-Ali et al.	N/A	N/A
2013/0331660	12/2012	Al-Ali et al.	N/A	N/A
2013/0331670	12/2012	Kiani	N/A	N/A
2013/0331778	12/2012	Kruse et al.	N/A	N/A
2013/0345921	12/2012	Al-Ali et al.	N/A	N/A
2014/0012100	12/2013	Al-Ali et al.	N/A	N/A
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2014/0051953	12/2013	Lamego et al.	N/A	N/A
2014/0066783	12/2013	Kiani et al.	N/A	N/A
2014/0077956	12/2013	Sampath et al.	N/A	N/A
2014/0081100	12/2013	Muhsin et al.	N/A	N/A
2014/0081175	12/2013	Telfort	N/A	N/A
2014/0100434	12/2013	Diab et al.	N/A	N/A
2014/0114199	12/2013	Lamego et al.	N/A	N/A
2014/0120564	12/2013	Workman et al.	N/A	N/A
2014/0121482	12/2013	Merritt et al.	N/A	N/A
2014/0127137	12/2013	Bellott et al.	N/A	N/A
2014/0129702	12/2013	Lamego et al.	N/A	N/A
2014/0135588	12/2013	Al-Ali et al.	N/A	N/A
2014/0142401	12/2013	Al-Ali et al.	N/A	N/A
2014/0163344	12/2013	Al-Ali	N/A	N/A
2014/0163402	12/2013	Lamego et al.	N/A	N/A
2014/0166076	12/2013	Kiani et al.	N/A	N/A
2014/0171763	12/2013	Diab	N/A	N/A
2014/0180038	12/2013	Kiani	N/A	N/A
2014/0180154	12/2013	Sierra et al.	N/A	N/A
2014/0180160	12/2013	Brown et al.	N/A	N/A
2014/0187973	12/2013	Brown et al.	N/A	N/A

2014/0213864	12/2013	Abdul-Hafiz et al.	N/A	N/A
2014/0266790	12/2013	Al-Ali et al.	N/A	N/A
2014/0275808	12/2013	Poeze et al.	N/A	N/A
2014/0275835	12/2013	Lamego et al.	N/A	N/A
2014/0275871	12/2013	Lamego et al.	N/A	N/A
2014/0275872	12/2013	Merritt et al.	N/A	N/A
2014/0276115	12/2013	Dalvi et al.	N/A	N/A
2014/0288400	12/2013	Diab et al.	N/A	N/A
2014/0309615	12/2013	Mazlish	N/A	N/A
2014/0316217	12/2013	Purdon et al.	N/A	N/A
2014/0316218	12/2013	Purdon et al.	N/A	N/A
2014/0316228	12/2013	Blank et al.	N/A	N/A
2014/0323825	12/2013	Al-Ali et al.	N/A	N/A
2014/0323897	12/2013	Brown et al.	N/A	N/A
2014/0323898	12/2013	Purdon et al.	N/A	N/A
2014/0330092	12/2013	Al-Ali et al.	N/A	N/A
2014/0330098	12/2013	Merritt et al.	N/A	N/A
2014/0330099	12/2013	Al-Ali et al.	N/A	N/A
2014/0336481	12/2013	Shakespeare et al.	N/A	N/A
2014/0357966	12/2013	Al-Ali et al.	N/A	N/A
2015/0005600	12/2014	Blank et al.	N/A	N/A
2015/0011907	12/2014	Purdon et al.	N/A	N/A
2015/0012231	12/2014	Poeze et al.	N/A	N/A
2015/0025406	12/2014	Al-Ali	N/A	N/A
2015/0032029	12/2014	Al-Ali et al.	N/A	N/A
2015/0038859	12/2014	Dalvi et al.	N/A	N/A
2015/0045637	12/2014	Dalvi	N/A	N/A
2015/0051462	12/2014	Olsen	N/A	N/A
2015/0073241	12/2014	Lamego	N/A	N/A
2015/0080754	12/2014	Purdon et al.	N/A	N/A
2015/0087936	12/2014	Al-Ali et al.	N/A	N/A
2015/0094546	12/2014	Al-Ali	N/A	N/A
2015/0097701	12/2014	Muhsin et al.	N/A	N/A
2015/0099950	12/2014	Al-Ali et al.	N/A	N/A
2015/0099951	12/2014	Al-Ali et al.	N/A	N/A
2015/0099955	12/2014	Al-Ali et al.	N/A	N/A
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2015/0106121	12/2014	Muhsin et al.	N/A	N/A
2015/0112151	12/2014	Muhsin et al.	N/A	N/A
2015/0116076	12/2014	Al-Ali et al.	N/A	N/A
2015/0126830	12/2014	Schurman et al.	N/A	N/A
2015/0133755	12/2014	Smith et al.	N/A	N/A
2015/0141781	12/2014	Weber et al.	N/A	N/A
2015/0165312	12/2014	Kiani	N/A	N/A
2015/0196237	12/2014	Lamego	N/A	N/A
2015/0216459	12/2014	Al-Ali et al.	N/A	N/A
2015/0230755	12/2014	Al-Ali et al.	N/A	N/A
2015/0238722	12/2014	Al-Ali	N/A	N/A
2015/0245773	12/2014	Lamego et al.	N/A	N/A
2015/0245794	12/2014	Al-Ali	N/A	N/A

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2015/0272514	12/2014	Kiani et al.	N/A	N/A
2015/0351697	12/2014	Weber et al.	N/A	N/A
2015/0351704	12/2014	Kiani et al.	N/A	N/A
2015/0359429	12/2014	Al-Ali et al.	N/A	N/A
2015/0366472	12/2014	Kiani	N/A	N/A
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2015/0374298	12/2014	Al-Ali et al.	N/A	N/A
2015/0380875	12/2014	Coverston et al.	N/A	N/A
2016/0000362	12/2015	Diab et al.	N/A	N/A
2016/0007930	12/2015	Weber et al.	N/A	N/A
2016/0029932	12/2015	Al-Ali	N/A	N/A
2016/0045118	12/2015	Kiani	N/A	N/A
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2016/0058347	12/2015	Reichgott et al.	N/A	N/A
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2016/0113527	12/2015	Al-Ali	N/A	N/A
2016/0143548	12/2015	Al-Ali	N/A	N/A
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2016/0166183	12/2015	Poeze et al.	N/A	N/A
2016/0166188	12/2015	Bruinsma et al.	N/A	N/A
2016/0166210	12/2015	Al-Ali	N/A	N/A
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Background/Summary

INCORPORATION BY REFERENCE TO ANY RELATED APPLICATIONS

(1) Any and all applications, if any, for which a foreign or domestic priority claim is identified in the Application Data Sheet of the present application are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

(2) Pulse oximetry is a widely accepted noninvasive procedure for measuring the oxygen saturation level of arterial blood, an indicator of a person's oxygen supply. A typical pulse oximetry system utilizes an optical sensor attached to a fingertip to measure the relative volume of oxygenated hemoglobin in pulsatile arterial blood flowing within the fingertip. Oxygen saturation (SpO₂), pulse rate and a plethysmograph waveform, which is a visualization of pulsatile blood flow over time, are displayed on a monitor accordingly.

(3) Conventional pulse oximetry assumes that arterial blood is the only pulsatile blood flow in the measurement site. During patient motion, venous blood also moves, which causes errors in conventional pulse oximetry. Advanced pulse oximetry processes the venous blood signal so as to report true arterial oxygen saturation and pulse rate under conditions of patient movement. Advanced pulse oximetry also functions under conditions of low perfusion (small signal amplitude), intense ambient light (artificial or sunlight) and electrosurgical instrument interference, which are scenarios where conventional pulse oximetry tends to fail.

(4) Advanced pulse oximetry is described in at least U.S. Pat. Nos. 6,770,028; 6,658,276; 6,157,850; 6,002,952; 5,769,785 and 5,758,644, which are assigned to Masimo Corporation ("Masimo") of Irvine, California and are incorporated in their entirety by reference herein. Corresponding low noise optical sensors are disclosed in at least U.S. Pat. Nos. 6,985,764; 6,813,511; 6,792,300; 6,256,523; 6,088,607; 5,782,757 and 5,638,818, which are also assigned to Masimo and are also incorporated in their entirety by reference herein. Advanced pulse oximetry systems including Masimo SET® low noise optical sensors and read through motion pulse oximetry monitors for measuring SpO₂, pulse rate (PR) and perfusion index (PI) are available from Masimo. Optical sensors include any of Masimo LNOP®, LNCS®, SofTouch™ and Blue™ adhesive or reusable sensors. Pulse oximetry monitors include any of Masimo Rad-8®, Rad-5®, Rad®-5v or SatShare® monitors.

(5) Advanced blood parameter measurement systems are described in at least U.S. Pat. No. 7,647,083, filed Mar. 1, 2006, titled Multiple Wavelength Sensor Equalization; U.S. Pat. No. 7,729,733, filed Mar. 1, 2006, titled Configurable Physiological Measurement System; U.S. Pat.

Pub. No. 2006/0211925, filed Mar. 1, 2006, titled Physiological Parameter Confidence Measure and U.S. Pat. Pub. No. 2006/0238358, filed Mar. 1, 2006, titled Noninvasive Multi-Parameter Patient Monitor, all assigned to Cercacor Laboratories, Inc., Irvine, CA (Cercacor) and all incorporated in their entirety by reference herein. Advanced blood parameter measurement systems include Masimo Rainbow® SET, which provides measurements in addition to SpO₂, such as total hemoglobin (SpHb™), oxygen content (SpOC™), methemoglobin (SpMet®), carboxyhemoglobin (SpCO®) and PVI®. Advanced blood parameter sensors include Masimo Rainbow® adhesive, ReSposable™ and reusable sensors. Advanced blood parameter monitors include Masimo Radical-7™, Rad-87™ and Rad-57™ monitors, all available from Masimo. Such advanced pulse oximeters, low noise sensors and advanced blood parameter systems have gained rapid acceptance in a wide variety of medical applications, including surgical wards, intensive care and neonatal units, general wards, home care, physical training, and virtually all types of monitoring scenarios.

SUMMARY OF CERTAIN EMBODIMENTS

(6) Hospital patients and out-patients may experience negative effects from various pharmaceutical drugs prescribed for treatment. These negative effects may be due to overdose, under dose or adverse reactions to these drugs. As one example, opioids are often given to post-surgical patients for pain management, and while opioid use is safe for most patients, opioid analgesics are often associated with adverse reactions, such as respiratory depression as a result of excessive dosing, improper monitoring, medication interactions and adverse reactions with other drugs.

Complications from opioid use are inevitable and not always avoidable. However, when a patient dies because such complications were not timely recognized or treated appropriately, it is termed a “failure to rescue.”

(7) Failure to rescue in the case of opioid overdose can be largely prevented by an advantageous autonomous administration of an opioid antagonist. Similarly, other serious patient conditions such as heart arrhythmia, high or low heart rate, high or low blood pressure, hypoglycemia and anaphylactic shock, to name a few, may be moderated by the autonomous administration of an appropriate rescue drug. An autonomous drug delivery system (ADDS) advantageously utilizes physiological monitor outputs so as to automatically give at least one bolus of at least one drug or medicine when certain criteria and confidence levels are met. In addition, an emergency button is provided to manually trigger administration of such a rescue drug or medicine. In an embodiment, the emergency button may be triggered remotely. ADDS advantageously responds with a rescue drug when delay in clinician response would otherwise result in patient injury or death.

(8) One aspect of an autonomous drug delivery system is a housing with a front panel and a drug compartment. A display and buttons are disposed on the front panel. An IV injector is disposed in the drug compartment, and a nozzle extending from the housing is in mechanical communications with the IV injector. In an embodiment, the IV injector has a drug reservoir, a piston partially disposed within and extending from the drug reservoir and a drive motor in mechanical communications with the piston. The drive motor actuates so as to drive the piston against a drug containing bolus disposed within the drug reservoir. The piston causes the drug to extrude from the nozzle.

(9) Another aspect of an autonomous drug delivery system is receiving a physiological parameter indicative of a medical state of a person and calculating a trigger condition and a corresponding confidence indicator. An alarm occurs if the trigger condition exceeds a predetermined threshold. A drug is administered to the person if the alarm is not manually acknowledged within a predetermined time interval.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-C are front, front partial-cutaway, and cutaway drug reservoir views of an autonomous drug delivery system (ADDS) embodiment;
- (2) FIG. 2 is a front view of an autonomous drug delivery system configured with a conventional IV drip;
- (3) FIG. 3 is a front view of an autonomous drug delivery system configured with an IV drip and a patient controlled analgesia (PCA) pump;
- (4) FIG. 4 is a block diagram of an autonomous drug delivery system having fluid connectivity to a patient and electronic communications with sensor(s) and monitor(s);
- (5) FIG. 5 is a single-rail drug administration decision flow chart;
- (6) FIG. 6 is a cutaway dual drug reservoir view of an autonomous drug delivery system embodiment;
- (7) FIG. 7 is a dual-rail drug administration decision flow chart; and
- (8) FIG. 8 is a block diagram of an autonomous drug delivery system signal processing and instrument management.
- (9) FIG. 9 is a block diagram of an example automated first responder system.
- (10) FIG. 10 is a block diagram of an example automated first responder process.
- (11) FIG. 11 is a block diagram of example diagnostic models.
- (12) FIG. 12 is a block diagram of another example automated first responder process.

DETAILED DESCRIPTION

(13) FIGS. 1A-C illustrate an autonomous drug delivery system (ADDS) **100** that inputs physiological parameters generated by one or more patient monitors, such as SpO₂, PR, RR, EtCO₂ and BP, and evaluates certain criteria and confidence levels based upon those parameters, so as to automatically give a bolus of a rescue drug. As shown in FIG. 1A, the autonomous drug delivery system **100** has a control section **102** and a drug section **108**. The control section **102** has a front panel **104** and an electrical interface (not visible). The front panel **104** provides a user interface including a user display **110**, such as an LED or LCD readout of device and patient status, among other data; control buttons **120** such as power on/off, reset, mode among other functions, and an emergency button **130** to manually trigger administration of the rescue drug. The emergency button **130** may have a flip-up cover or other safety mechanism, such as a simultaneous press with another button, so as to prevent inadvertent drug administration. The electrical interface communicates with patient monitor(s), alarm(s) and the drug section **108**. The drug section **108** has a cover **109** concealing a drug administration device installed within a recessed compartment, described below.

(14) As shown in FIG. 1B, the cover **109**, once flipped-up, slid-open, removed or otherwise opened, exposes an IV injector **150** having a drug reservoir **170**, a piston **180** and a drive motor **190** disposed within a drug compartment **152**. The drug reservoir **170** is configured to accept a disposable bolus of a rescue drug. When actuated, the motor **190** drives the piston **180** into the drug reservoir **170** so that the rescue drug contained in the drug reservoir **170** is expelled out of the nozzle **178** (FIG. 1C) and into an attached IV tube that carries the rescue drug to the patient, as described with respect to FIGS. 2-3, below. The bolus remains in the drug reservoir **170** until it is used or expires. The expiration date can optionally be read from the bolus or otherwise entered into the ADDS memory (part of the DSP **810** FIG. 8)

(15) As shown in FIG. 1C, in a ready position, the drug reservoir **170** has a generally-cylindrical enclosure **172** containing an unused rescue drug **174** and a piston **180**. In particular, the piston **180** has a piston head **182** disposed within the enclosure **172** and a piston shaft **184** extending from the enclosure **172**. The motor acts on the piston shaft **184** so as to move the piston head **182** from a first position (shown) distal the nozzle **178** to a second position proximate the nozzle, thus expelling the rescue drug from the reservoir **170** and into an IV tube (not shown) attached to the

nozzle **178**.

(16) FIG. 2 illustrates a conventional IV drip configuration **200** incorporating an autonomous drug delivery system **100**. In particular, an IV pole **10** mounts an IV bag **20** in fluid communications with a patient **1** via an IV line **30**. Splicing the IV line is a one-way valve **50** in communications with the ADDS **100** via a line shunt **40**. In the event the ADDS **100** is triggered, either manually via the emergency button **130** (FIG. 1A) or automatically via patient monitor inputs in communications with the control section **102** (FIG. 1A), the ADDS **100** injects its rescue drug **174** (FIG. 1C) into the shunt **40** and into the IV line **30** via the one-way valve **50**, which shuts off the IV bag drip **20** blocking further IV bag **20** fluid.

(17) FIG. 3 illustrates another conventional IV drip configuration **300** incorporating an autonomous drug delivery system **100** in addition to a patient controlled analgesia (PCA) pump **70**. Similar to the FIG. 2 configuration described above, an IV pole **10** mounts an IV bag **20** in fluid communications with a patient **1** via an IV line **30**. Splicing the IV line is a one-valve **50** in communications with the rescue device **100** via a line shunt **40**. Also splicing the IV line is a one-valve **60** in communications with the PCA pump. As described above, in the event the ADDS **100** is triggered, either manually via the emergency button **130** (FIG. 1A) or automatically via patient monitor inputs in communications with the control section **102** (FIG. 1A), the ADDS **100** injects its rescue drug **174** (FIG. 1C) into the shunt **40** and into the IV line **30** via the one-way valve **50**, which shuts off the IV bag drip **20** and the PCA analgesia **72**.

(18) The ADDS **100** may be used for a variety of medical emergencies and conditions in conjunction with a variety of rescue drugs. In one embodiment, the ADDS is advantageously used in conjunction with a patient controlled analgesia (PCA) device that delivers pain medication upon patient demand. For example, a patient can push a button to request a morphine injection. Here, a patient is particularly vulnerable to inadvertently overdosing themselves to the point of respiratory failure. The ADDS inputs one or more parameters such as respiration rate, oxygen saturation, pulse rate, blood pressure and end-tidal CO₂ according to sensors attached to the patient and corresponding monitors and delivers an opioid antagonist such as Naloxone (Narcan), as described in further detail with respect to FIG. 4, below.

(19) The ADDS **100** can optionally be responsive to heart arrhythmias and deliver a rescue drug specific to the type of rhythm detected; can optionally be responsive to a high heart rate or low heart rate so as to deliver a heart rate regulating drug; can optionally be responsive to high or low blood pressure so as to deliver a blood pressure regulating drug; can optionally be responsive to hypoglycemia so as to deliver glucagon; or can be responsive to other specific conditions and deliver appropriate medicine. In various embodiments, the ADDS **100** delivers compounded (mixed) formulas of more than one drug. This is accomplished through a mixture of drugs stored in a single reservoir or the use of multiple reservoirs. In all of these embodiments, the response is always a bolus (reservoir) administration. The drug is infused in a single administration step all at once (as opposed to over a prescribed amount of time as in a pump). The ADDS is capable of delivering multiple bolus doses, where subsequent bolus doses are ideally enabled after manual review of the initial, automatically delivered dose.

(20) The ADDS can be configured to administer the bolus at a specific date and time, rather than in response to a monitored condition. The bolus administration can be at multiple discrete times or over a specified time interval. When the ADDS is responsive to multiple physiological parameters received from one or more monitoring instruments, a prescribed minimum and or maximum interval from a previous dose or start time may be used.

(21) In various situations, physiological “guard rails” must be maintained. (See, for example FIGS. 6-7). As an example, a hypotensive drug may be used to raise a patient's blood pressure during emergencies. As another example, a drug to regulate heart rhythm or suppress arrhythmias may be used. In an embodiment, the display **110** (FIG. 1A) indicates the device is working and receiving proper inputs. The ADDS **100** has a memory that records data that can be recalled to indicate a

trigger cause for administering a rescue drug, such as the time and amount of drug administered and other logged conditions. In an embodiment, the rescue drug is administered in a progression rather than all-at-once. For example, a 10% dose may be given and the patient monitored for improvement, with additional doses spaced at intervals as needed. In an embodiment, a spring driven piston with a releasable catch is used to administer the rescue drug in lieu of a motor-driven piston. In an embodiment, an ADDS is advantageously integrated with a PCA device as a single instrument.

(22) FIG. 4 illustrates an autonomous drug delivery system **400** having fluid connectivity to a patient **412** and electronic communications **424** with one or more monitors **420**. One or more sensors **430** interface **432** with the patient **412**, either by direct patient attachment or by remote sensing. For example, an optical sensor **430** may clip to a fingertip site **432** and provide pulsatile blood flow data via cable **422** to a pulse oximeter or blood parameter monitor **420**. The pulse oximeter or blood parameter monitor **420** calculates physiological parameters **424**, such as oxygen saturation and pulse rate, which are transmitted to the ADDS **410**. The ADDS **410** responds to the parameters **424** by regulating (starting, increasing, decreasing or halting) the drug dosage **412** accordingly. The ADDS **410** may generate one or more alarms **414** to alert a caregiver. The caregiver may reset **416** an alarm upon responding to the alert. In an embodiment, communication between the ADDS **410** and a caregiver may be via a wired or wireless network and received at a nurses' station or equivalent.

(23) FIG. 5 illustrates example single-rail drug administration decisions. In this case, there is a single threshold (maximum or minimum) that if exceeded, triggers the ADDS device to administer a rescue drug to a patient. One or more parameters **501** are input to the ADDS device. An ADDS algorithm **510** calculates a trigger condition and an associated confidence indicator **512**. If the ADDS trigger threshold is exceeded **520**, an alarm **530** is triggered **522**. Otherwise **524**, ADDS does nothing. If the alarm **532** is timely acknowledged **540**, such as by a caregiver pressing an ADDS acknowledge button **544**, then ADDS exits **560** and the alarm is silenced. If the alarm **532** is not acknowledged **542**, then ADDS administers a rescue drug **550**. Note that ADDS has a manual trigger **570** (button press), so as to manual trigger **572** drug administration **550** (see **130** FIG. 1A). In an embodiment, alarms **530** and corresponding acknowledgment times **540** may progress from a local alarm, a network alarm and finally a pager alarm before autonomous drug delivery is triggered.

(24) FIG. 6 illustrates a dual drug reservoir embodiment **600** of an autonomous drug delivery system. A first drug reservoir **670** and a second drug reservoir **680** may each contain the same drug for one or two separate doses or may contain different drugs for different single doses depending on the monitored condition. See, for example FIG. 7. The drug reservoirs **670**, **680** share a common nozzle **662**. The nozzle **662** has a junction **660** that connects fluid tubes **678**, **688** to respect first **676** and second **686** reservoir nozzles. In an embodiment, a clinician has to re-enable the ADDS before a second dose is administered.

(25) FIG. 7 illustrates dual-rail drug administration decisions. In this case, there is a maximum threshold and a minimum threshold. If the maximum threshold is exceeded, for example, one or more parameter values or combinations of parameter values goes over a predetermined limit, then the ADDS device is triggered to administer a rescue drug to a patient. Also, If the minimum threshold is exceeded, for example, one or more parameter values or combinations of parameter values falls below a predetermined limit, then the ADDS device is triggered to administer a rescue drug to a patient. In an embodiment, one rescue drug may be administered upon exceeding the maximum threshold and a different rescue drug may be administered upon exceeding the minimum threshold. One or more parameters **701** are input to the ADDS device. ADDS algorithms **710** and **760** calculate maximum (HI) trigger conditions and minimum (LO) trigger conditions and associated confidence indicator. If either the HI or LO ADDS trigger thresholds are exceeded **720**, **770** the corresponding alarm **730**, **780** is triggered. Otherwise **724**, **774** ADDS does nothing. If a HI

or LO alarm **730, 780** is timely acknowledged, such as a caregiver pressing an ADDS acknowledge button **744, 794**, then ADDS exits **741, 791** and the alarm is silenced. If the alarm **730, 780** is not acknowledged **742, 792** then ADDS administers the appropriate HI rail **745** or LO rail **795** rescue drug. Note that ADDS has manual triggers **799** (LO or HI button presses), so as to manually trigger either the LO rail drug **745** or the HI rail drug **795**.

(26) FIG. **8** illustrates an autonomous drug delivery system (ADDS) controller **800** having a digital signal processor (DSP) **810**, an instrument manager **820** and interfaces **830**. The DSP **810** has algorithms, such as described with respect to FIGS. **5-7**, above, for determining trigger conditions, alarms and trigger timing **812**. The instrument manager **820** interfaces with the DSP **810** so as to drive the ADDS displays **832** and alarms **834**, to receive inputs from the buttons/keypad **835**, to communicate with external monitors **837** that derive triggering parameters from patient sensors, to control the injector motor(s) and read injector status **839**, such as bolus full/empty and injector active (delivering rescue drug). In an embodiment, DSP firmware records in memory the time and duration of events that triggered drug delivery, including a before and after history of monitor parameters that triggered drug administration.

Additional Embodiments

(27) The following additional embodiments can be implemented together with any of the devices, algorithms, or features described above, in any combination.

(28) FIG. **9** depicts an example automated first responder system **900**. In the automated first responder system **900**, a patient **902** is shown in communication with a patient monitor **910** and an autonomous drug delivery device **920**. For instance, the patient monitor **910** may be in wireless or electrical communication with one or more sensors coupled with the patient. Examples of such sensors are described above, and may include, for example, optical sensors, acoustic respiratory sensors, ECG sensors, brainwave measurement sensors, blood pressure sensors, and the like.

(29) The patient monitor **910** and the autonomous drug delivery device **920** can have some or all of the features of similar devices described above. For instance, the patient monitor **910** can include any of the controller features described above with respect to FIG. **8**. In general, the patient monitor **910** may include one or more hardware processors, memory, and a display. Alternatively, the display may be optional, and the patient monitor **910** can output patient data to a separate display. Likewise, the autonomous drug delivery device **920** can have some or all the features of the ADDS embodiments described above.

(30) The patient monitor **910** is shown in communication with a gateway device **940**. The patient monitor **910** may communicate with the gateway device **940** wired or wirelessly over a hospital network (not shown). The hospital network may include a local area network, a wide area network, an intranet, the Internet, combinations of the same, or the like. The gateway device **940** can facilitate communication across the network between multiple devices shown in the system **900**. These devices can include clinician devices **960**, which may include tablets, pagers, smart phones, laptops, computers, nurses' station computers, and the like. In addition, the gateway **940** is shown in communication with an electronic medical record (EMR) database **950**, which can store lab results **930** obtained from laboratory tests applied to the patient **902**. The patient monitor **910** may access the EMR **950** through the gateway **940** to obtain access to the lab results **930** as well as other data, including data regarding charting of the patient's conditions, medications, tests, and the like applied during a hospital stay. Of course, the system **900** can also be applied in any clinical setting, and not just in a hospital. However, for convenience, this specification refers specifically to hospital embodiments, without loss of generality.

(31) The automated first responder system **900** can implement any of the features described above with respect to FIGS. **1-8**. In addition, in certain embodiments, the automated first responder system **900** can also implement the additional features discussed below. For instance, the patient monitor **910** can initiate treatment workflows automatically based on an analysis of patient data. The patient data can include data measured from physiological sensors, as well as data obtained

from lab results and the like. Using this patient data, the patient monitor **910** can determine whether the data corresponds with or matches any diagnostic model or models stored in physical computer storage. A diagnostic model can be represented as a data structure or the like and can relate a set of patient data inputs to a possible diagnosis and to a treatment workflow to address the possible diagnosis. Example treatment workflows are described below with respect to FIG. **11**.

(32) If the patient data matches a diagnostic model (examples of which are described below), the patient monitor **910** can initiate a treatment workflow automatically to begin treatment of the patient. The treatment workflow can include administration of a substance, such as a drug, using any of the techniques described above. The treatment workflow can also include other actions, such as ordering lab tests, alerting clinicians, and the like.

(33) As part of a treatment workflow, the patient monitor **910** may communicate an instruction or signal to the autonomous drug delivery device **920** to cause the autonomous drug delivery device **920** to supply a dose or bolus of medication to the patient **902**. In one embodiment, the autonomous drug delivery device **920** is an intravenous therapy device, such as an intravenous (IV) drip device. Thus, the autonomous drug delivery device **920** may administer substances other than drugs to the patient **902**, such as saline solution. In some embodiments, a saline solution may be combined with a drug to be administered to the patient **902**.

(34) As another example aspect of a treatment workflow, the treatment workflow may include initially administering a first dosage of a substance to a patient without requiring initiation by a clinician. Optionally, however, the patient monitor **910** may produce an output (for example, on the display) that provides the clinician with an option to intervene and stop administration of a substance or stop any other aspect of a treatment workflow.

(35) Later, if the patient data again matches the diagnostic model, such that an additional administration of a substance may be warranted, the treatment workflow may include alerting a clinician instead of administering the second dosage. In this manner, it is possible to administer a relatively harmless or low dosage of a substance first to the patient without human intervention in order to provide an automated first response but subsequently to involve a clinician to avoid accidentally overdosing the patient with the substance.

(36) As another example, the patient monitor **910**, upon detecting that a diagnostic model has been matched, can alert a clinician who may be remote from the patient of the patient's potential condition. The clinician can then respond to the alert in various ways, discussed below.

(37) The alert to the clinician may be sent as a message over a network, such as the hospital network, to a clinician's device (such as a smartphone, tablet, laptop, desktop, or other computing device). The alert may be sent as an electronic mail message, a text message, or as another type of message, including a custom message for a custom mobile application or web application installed or accessed on the clinician's device. The alert can include any subset of the following information: patient identifying and/or demographic information, the patient data or a summary thereof (such as waveform(s) and/or physiological parameter value(s)), an indication of the patient's condition such as an indication that a specific diagnostic model has been met (such as "this patient may have sepsis" or "possible opioid overdose detected"), and a recommended treatment course (such as a recommendation to administer a bolus of medication, order a lab test, or any other aspect of a workflow discussed herein).

(38) Upon receipt of the alert, software on the clinician's device can output the message to the clinician optionally with functionality for the clinician to respond. For instance, the clinician's device may have a mobile application or browser installed that can receive the alert and output a user interface with the alert for presentation to the clinician. The user interface may include buttons, input fields, or other user interface controls that allow the clinician to interact with the alert. For instance, the user interface may provide functionality for the clinician to accept or adjust the recommended treatment. Should the clinician accept the recommended treatment, the software on the clinician's device can transmit the acceptance to the patient monitor **910**, which can cause

the autonomous drug delivery device **920** to deliver the recommended dosage to the patient (or order a lab or perform another workflow action). Should the clinician adjust the recommended treatment, this adjustment may be communicated to the patient monitor **910**, which can cause the autonomous drug delivery device **920** to deliver the adjusted dosage to the patient (or order a lab or perform another workflow action).

(39) Likewise, the clinician can remotely create or adjust a treatment workflow, whether lab order or drug administration or other workflow, and can cause this adjustment to be sent to the patient monitor **910** for application to the patient. For example, the clinician may receive patient data with or without a recommendation for treatment. The clinician can then access a user interface on the clinician's device to input a desired treatment or workflow, even if one was not recommended by the patient monitor **910**. The clinician device can transmit this selected treatment or workflow to the patient monitor **910**. If the treatment or workflow includes the administration of a drug, the patient monitor **910** can cause the autonomous drug delivery device **920** to deliver the selected dosage to the patient. Any of the autonomous drug delivery devices described herein can include a plurality of different intervention medications ready to be injected or supplied to the patient intravenously as desired.

(40) FIG. **10** depicts an example automated first responder process **1000**. The automated first responder process **1000** can be implemented by any of the systems described above. For instance, the patient monitor **910** can implement the process **1000**. In another embodiment, however, at least part of the process **1000** can be implemented in any server, for instance, in a cloud computing environment. For example, the process **1000** can be implemented in a server that receives information from the patient monitor **910**, and which provides outputs to the monitor **910** to enable the monitor **910** to perform portions of a treatment workflow (such as initiate delivery of substance by the autonomous drug delivery device **920**). For ease of description, however, the process **1000** is described as being implemented primarily by the patient monitor **910**.

(41) At block **1002**, the patient monitor **910** receives patient data including one or more physiological parameter values. The one or more physiological parameter values may include values of one or more physiological parameters, such as respiration rate, oxygen saturation, blood pressure, heart rate, and the like (as well as any other parameters discussed above). In addition to receiving one or more physiological parameter values, the patient monitor may receive additional patient data, such as lab test values. One example of a lab test value that may be useful is white blood cell count (WBC). More generally, any blood test value, urine test value, stool test value, or any other lab test value may be used.

(42) At block **1004**, the patient monitor **910** analyzes patient data based on one or more diagnostic models. For example, the patient monitor **910** can compare the patient data to any diagnostic models stored in memory or other physical computer storage. The diagnostic models may include one or more criteria for determining whether to initiate a treatment workflow. Evaluating this criteria can involve performing a comparison of the patient data with specific values, thresholds, trends, changes, combinations, ratios, and the like. A specific combination of parameter trends, for instance, may indicate that a patient is likely experiencing an opioid overdose, sepsis, hypovolemia, or any of a number of other conditions. Specific examples of diagnostic models are presented below with respect to FIG. **11**.

(43) At block **1006**, the patient monitor **910** determines whether the patient data matches a diagnostic model. If not, the process **1000** loops back to block **1002**, and the patient monitor **910** continues to receive patient data. If the patient monitor **910** identifies a match in a diagnostic model, however, the process **1000** proceeds to block **1008**, where the patient monitor **910** determines whether this is the first time that the patient data matches this model. If so, at block **1010** the patient monitor **910** initiates a treatment workflow specific to the diagnostic model. The process **1000** then loops back to block **1002**.

(44) Diagnostic models can be represented as data structures or data definitions stored in physical

computer storage or memory. Diagnostic models can also be generated empirically by collecting data from a plurality of patients, measuring the same or similar physiological parameters, and by analyzing patient treatments and subsequent outcomes. Diagnostic models can also be rule-based models, or a combination of empirical and rule-based models. Some example rule-based models are described below with respect to FIG. 11.

(45) In some embodiments, at block **1006**, the patient monitor **910** calculates a confidence value that represents a likelihood that the patient data matches the diagnostic model. The confidence value can be a percentage value, a score on any scale, or the like. The patient monitor **910** can calculate a higher confidence value based on the patient data more closely matching the diagnostic model or a lower confidence value based on the patient data less closely matching the diagnostic model.

(46) The confidence value can also be based determined at least in part on parameter value confidence. For instance, the patient monitor **910** can calculate a confidence value for a parameter value, indicating a likelihood of the parameter value being accurate. The confidence value for matching the diagnostic model can therefore take into account parameter confidence values. For example, if two parameters are analyzed with respect to a diagnostic model, and the two parameters have a calculated confidence of 95% and 75% respectively, the patient monitor **910** can calculate a confidence value that is the mean of these two values (85%) or some other combination of these two values (such as selecting the lower confidence value as the overall confidence value). Then, the patient monitor **910** can compare the two parameter values with the diagnostic model and determine how closely the parameters match the model. If the parameters match the model well, the patient monitor **910** can give a high confidence score, but the patient monitor **910** may then modify this high confidence score based on the parameter value confidence scores. For instance, if the confidence value for matching the model is 95%, but the parameter value confidence is 75%, the patient monitor **910** may lower the confidence for matching the model (e.g., by multiplying the two numbers to reach ~71%). Many other ways for calculating confidence are possible.

(47) The patient monitor **910** can determine that if the confidence value is too low, that the diagnostic model is not met. Alternatively, the patient monitor **910** can determine that the diagnostic model is met even when a confidence value is low but may output an indication of the confidence value (for example, on a display) so that a clinician can determine whether to trust the diagnostic model match. The patient monitor **910** can output an indication of the confidence value regardless of the value, whether it be high or low. The indication can be the value itself or some representation of that value, such as a graphic representation (for example, a bar with a size corresponding to the confidence value).

(48) However, if at block **1008** it is not the first time that the patient data matches this model, then at block **1012** the patient monitor **910** alerts a clinician so that the clinician can determine whether to give an additional substance or perform an additional action associated with the treatment workflow. The patient monitor **910** can alert the clinician by outputting the alert or alarm audibly and/or visually. The patient monitor **910** can also alert the clinician by transmitting an alert or alarm message to the clinician over a network, such as a hospital network, a local area network (LAN), a wide area network (WAN), the Internet, an Intranet, or the like.

(49) Thus, the process **1000** can provide an automated treatment of the patient as a first response and defer clinician intervention until a subsequent response. For instance, a first time when it is determined that the patient data indicates a possible opioid overdose, the patient monitor can cause an initial dosage of an overdose-treating medication like Narcan to be supplied to the patient. However, if later the patient data again indicates that the patient is suffering from a possible overdose, the patient monitor **910** can alert a clinician to determine whether the patient needs an additional dose.

(50) In an embodiment, if the clinician does not respond rapidly enough (for example, within a predetermined time period), the patient monitor **910** can automatically supply a second dose to the

patient. The patient monitor **910** may be restricted from supplying a second dose to the patient unless the patient data includes a life-threatening parameter value or values (such as a very low pulse rate, oxygen saturation value, or blood pressure). More generally, the patient monitor **910** can take into account any of the following variables when determining whether to administer a subsequent dose of a drug to a patient, among others: time elapsed since an alarm without a clinician disabling the alarm or otherwise responding to the alarm; severity of the patient data; amount of the earlier and/or subsequent dosage (a lower dose may be less harmful, and thus subsequent low doses might be permitted without clinician intervention); the type of drug administered (some drugs may have less risk for administering subsequent doses than others); specific facts about the patient including demographics, comorbidities, or the like (for example, a neonate might be at higher risk than an adult for a subsequent dose, or a patient with a certain disease might be at a higher risk than patients without that disease for a certain dose); other medications taken by the patient (a subsequent dose might be potentially more harmful if a patient is taking other medication that can react adversely upon a subsequent dose); prior permission by a clinician (the patient monitor **910** can output a user interface that enables a clinician to explicitly give permission for one or more subsequent automatic administration of doses when needed); combinations of the same, or the like.

(51) FIG. **11** depicts example diagnostic models **1110**, **1120**, and **1130**. These diagnostic models show example criteria corresponding to possible diagnoses and associated example treatment workflows. The first model **1110** is an opioid overdose model. In this model, if respiratory rate, heart rate, and blood pressure have decreased (for example, are trending downward or pass below a threshold), then these parameter changes may indicate an opioid overdose and may trigger a treatment workflow. A decrease in temperature can also be indicative of opioid overdose in the model. The example treatment workflow shown includes an administration of Narcan and informing a clinician. Informing a clinician may include sending an alert or alarm to clinician as described above, recording an indication in the patient's chart (such as in the EMR), combinations of the same or the like. The treatment workflow shown may be modified to include more or fewer steps. In addition, the steps of the treatment workflow and other treatment workflows described herein may be implemented in any order.

(52) The second model **1120** is a hypovolemia model. In this model **1120**, if blood pressure has decreased (for example, trending downward or below a threshold), heart rate is increased (for example, trending upward or above a threshold), and hemoglobin level (whether measured noninvasively (SpHb) or in a lab (THb)) is steady (for example, no positive or negative trend, or slope of the average trend is within a tolerance, or the parameter value has not changed more than a threshold amount for a period of time), this may be an indication that the patient is suffering from hypovolemia. The treatment workflow that may be triggered may include increasing a saline infusion rate (to increase blood volume and/or sodium levels in the blood), supplying oxygen (not shown; can increase the efficiency of the patient's remaining blood supply), and informing a clinician (similar to above). Although not shown, other steps may be included in these and other treatment workflows, such as ordering other drugs, recommending possible surgery (for example, to stop internal bleeding for some hypovolemic patients), and so on.

(53) The third model **1130** shown is a sepsis model. In this model, if blood pressure is down, white blood cell count is up, and temperature is up (for example, indicating a fever or conversely, too low temperature), the patient may be suffering from sepsis. Increased heart rate (for example, above a threshold), increased respiratory rate (for example, above a threshold), and decreased systolic blood pressure (for example, below a threshold) are other example factors that may be included in the model. The example treatment workflow shown includes activating a sepsis team (for example, informing a team of clinicians having different tasks that can assist with treating sepsis), administering drugs, and ordering labs, drugs, and radiology tests. The treatment workflow can also include automatically administering intravenous fluids and/or antibiotics. The treatment workflow

can also include recommending transfer of the patient to an intensive care unit (ICU).

(54) As stated above, there are many other diagnostic models that may be employed by the patient monitor to initiate treatment workflow. Any of the models may be combined or modified by other criteria. For instance, administration of some drugs can cause side effects, which may be detected or inferred from the patient data. As an example, administration of some drugs can impact blood pressure negatively. Detecting a change in blood pressure after a drug has been administered may cause a treatment workflow to begin that stops administration of the drug (if being administered automatically) or that recommends to a clinician to consider stopping administration of the drug.

(55) As another example, a diagnostic model may be provided for detecting bedsores formation (or risk factors for bedsores formation). This diagnostic model may take input from a pressure mat on the patient's bed or on the floor next to the bed, which can provide an electrical signal output when a patient moves or steps on the mat. Lack of movement or lack of stepping on a mat may indicate that the patient is not moving enough and is therefore susceptible to a bedsores. A possible treatment workflow may be executed, including alerting a caregiver that the patient should be moved in some manner to attempt to prevent bedsores formation.

(56) Diagnostic models may also take into account more than the physiological parameters and lab tests described herein as examples of patient data. Additional patient data may be relevant to evaluating whether a particular model is applicable. For instance, data regarding the patient's gender, age, or other comorbidities may impact whether or not the patient monitor determines that a particular diagnostic model is applicable. Likewise, any of these factors may be used to modify a treatment workflow when a diagnostic model is indicated as being applicable. For example, if a patient is indicated as being a smoker (which may be a data point stored in the EMR record for that patient), a treatment workflow that involves applying ventilation from a ventilator may include supplying more carbon dioxide than would be supplied to a non-smoking patient because the smoker's lungs are adapted to higher carbon dioxide levels when breathing.

(57) In another embodiment, a previous or current treatment may be an input into a diagnostic model as part of the patient data. Thus, a previous or current treatment may affect whether a diagnostic model is applied and/or how a treatment workflow is executed. For instance, if the patient has just had hip replacement surgery (which may be a data point stored in the EMR record for that patient), the patient monitor may be on heightened alert to search for indications of sepsis. The patient monitor may therefore be more sensitive to changes in blood pressure, temperature, and white blood cell count to determine whether sepsis is occurring. More specifically, a lower change in blood pressure, white blood cell count, or temperature than for other patients not having had hip replacement recently may be applied to trigger a sepsis treatment workflow.

(58) Moreover, as seen in the example workflow shown in FIG. 11, the level of alert provided to a clinician may vary depending on the seriousness of the diagnostic model detected. With sepsis, for instance, an entire team may be alerted, whereas with other conditions, like hypovolemia, a single clinician may be alerted. Of course, the number of clinicians alerted for any given condition can depend on the severity of that condition, the severity of the physiological parameter values, and so on, and may deviate from that shown in FIG. 11.

(59) Patient data in any of the models described herein can be analyzed according to a variety of different types of threshold values. Two general types of thresholds may be used—static thresholds and dynamic thresholds. Static thresholds can include predetermined thresholds that are the same for a class of patients, such as the same set of thresholds for all adults or a different set of thresholds for all neonates. Dynamic thresholds can include thresholds that are based on the current patient's mean or baseline physiological parameter values. Table 1 below depicts example static thresholds and dynamic thresholds for adult patients for some physiological parameters:

(60) TABLE-US-00001 TABLE 1 Example Adult Parameter Threshold Values

Physiological Parameter	Static Threshold	Dynamic Threshold
Low Respiratory Rate	≤ 6 RPM	$> 2SD$ from individual mean
Low Heart Rate	≤ 35 BPM	$> 2SD$ from individual mean
Low Blood Pressure	≤ 63	

mmHg >2SD from individual mean (MAP) Increased Heart Rate ≥ 90 BPM >2SD from individual mean Increased WBC $\geq 12,000/\mu\text{L}$ >2SD from individual mean Increased Temperature ≥ 38.3 degrees C. >2SD from individual mean

In Table 1, “SD” refers to “standard deviation,” “MAP” refers to mean arterial pressure, “RPM” refers to respirations or breaths per minute; and “BPM” refers to beats per minute.

(61) The static threshold category may be further subdivided into two subcategories—critical thresholds and normal thresholds. Critical thresholds may represent thresholds below (or above) which a physiological parameter is at a critical or life-threatening value. Normal thresholds can represent less-critical or less life threatening values that still nevertheless may be indicative of a diagnosis per a diagnostic model. Some diagnostic models can use critical thresholds, while others may use normal thresholds. Some diagnostic models may use a critical threshold for one parameter and a normal threshold for another parameter.

(62) The patient monitor can evaluate static thresholds or dynamic thresholds to determine whether a diagnostic model is met. In addition, the patient monitor can evaluate both static and dynamic thresholds to determine whether a diagnostic model is met. For instance, the patient monitor can perform a logical “OR” operation based on a comparison of physiological models to both static and dynamic thresholds. To illustrate one example implementation, if a patient's heart rate satisfies either a static or a dynamic threshold (or both), then the patient monitor can determine that the patient's heart rate satisfies a heart rate portion of a diagnostic model. In another embodiment, the patient monitor performs a logical “AND” operation (or some other logical operation) based on a comparison of physiological models to both static and dynamic thresholds.

(63) In an implementation, the patient monitor can rely on static thresholds early in a patient's hospital stay, where mean or baseline values of the patient's parameters are unknown or less reliable to calculate based on lack of patient data. Later in the patient's hospital stay (possibly even some minutes or hours after admission), the patient monitor can rely on dynamic thresholds after enough data about the patient is available to determine mean or baseline values.

(64) FIG. 12 depicts another example automated first responder process **1200**. The automated first responder process **1200** can be implemented by any of the systems described above. For instance, the patient monitor **910** can implement the process **1200**. In another embodiment, however, at least part of the process **1200** can be implemented in a server, for instance, in a cloud computing environment. The process **1200** can optionally be implemented in a server that receives information from the patient monitor **910**, and which provides outputs to the monitor **910**, to enable the monitor **910** to perform portions of a treatment workflow. For ease of description, however, the process **1200** is described as being implemented primarily by the patient monitor **910**.

(65) Blocks **1202** through **1206** can proceed the same or similar to blocks **1002** through **1006** of the process **1000** (see FIG. 10). At block **1208**, if the patient data matched a diagnostic model in block **1206**, the patient monitor **910** outputs an alarm or alert to a clinician. This alarm or alert can be provided in a similar manner to the alarm or alert output in block **1012** of the process **1000** (see FIG. 10). However, in this process **1200**, the patient monitor **910** determines at block **1210** whether the alarm (or alert) has been acknowledged within a predetermined time period (such as 30 seconds, 1 minute, 2 minutes, or some other time period). For instance, the patient monitor **910** can determine whether a clinician disabled the alarm as an indication that the alarm was acknowledged. Disabling of the alarm can indicate that the clinician visited the patient's bedside and explored the situation that led to the alarm.

(66) If the alarm was acknowledged within the predetermined time period, the process **1200** can end. If the alarm was not acknowledged within the predetermined time period, then at block **1212**, the patient monitor **910** can initiate a treatment workflow specific to the diagnostic model, as explained above with respect to block **1010** of FIG. 10.

(67) This process **1200** therefore enables the patient to receive some form of medical attention, such as specified by a treatment workflow, if a clinician is unable to treat the patient within a

certain amount of time. This process **1200** can save lives and increase patient comfort in busy hospitals or in other triage situations.

(68) Of course, should the patient data again match a diagnostic model, the patient monitor **910** can implement blocks **1008** and **1012** of the process **1000** or the like.

CONCLUSION

(69) An autonomous drug delivery system and an automated first responder system have been disclosed in detail in connection with various embodiments. These embodiments are disclosed by way of example only and are not to limit the scope of this disclosure or the claims that follow. One of ordinary skill in art will appreciate many variations and modifications. For instance, many other variations than those described herein will be apparent from this disclosure. For example, depending on the embodiment, certain acts, events, or functions of any of the algorithms described herein can be performed in a different sequence, can be added, merged, or left out altogether (for example, not all described acts or events are necessary for the practice of the algorithms).

Moreover, in certain embodiments, acts or events can be performed concurrently, for example, through multi-threaded processing, interrupt processing, or multiple processors or processor cores or on other parallel architectures, rather than sequentially. In addition, different tasks or processes can be performed by different machines and/or computing systems that can function together.

(70) The various illustrative logical blocks, modules, and algorithm steps described in connection with the embodiments disclosed herein can be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. The described functionality can be implemented in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the disclosure.

(71) The various illustrative logical blocks and modules described in connection with the embodiments disclosed herein can be implemented or performed by a machine, such as a hardware processor comprising digital logic circuitry, a general purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor can be a microprocessor, but in the alternative, the processor can be a controller, microcontroller, or state machine, combinations of the same, or the like. A processor can include electrical circuitry configured to process computer-executable instructions. In another embodiment, a processor includes an FPGA or other programmable device that performs logic operations without processing computer-executable instructions. A processor can also be implemented as a combination of computing devices, for example, a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. A computing environment can include any type of computer system, including, but not limited to, a computer system based on a microprocessor, a mainframe computer, a digital signal processor, a portable computing device, a device controller, or a computational engine within an appliance, to name a few.

(72) The steps of a method, process, or algorithm described in connection with the embodiments disclosed herein can be embodied directly in hardware, in a software module stored in one or more memory devices and executed by one or more processors, or in a combination of the two. A software module can reside in RAM memory, flash memory, ROM memory, EPROM memory, EEPROM memory, registers, hard disk, a removable disk, a CD-ROM, or any other form of non-transitory computer-readable storage medium, media, or physical computer storage known in the art. An example storage medium can be coupled to the processor such that the processor can read

information from, and write information to, the storage medium. In the alternative, the storage medium can be integral to the processor. The storage medium can be volatile or nonvolatile. The processor and the storage medium can reside in an ASIC.

(73) Conditional language used herein, such as, among others, “can,” “might,” “may,” “for example,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Further, the term “each,” as used herein, in addition to having its ordinary meaning, can mean any subset of a set of elements to which the term “each” is applied.

(74) Disjunctive language such as the phrase “at least one of X, Y and Z,” unless specifically stated otherwise, is to be understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z, or a combination thereof. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y and at least one of Z to each be present.

(75) Unless otherwise explicitly stated, articles such as “a” or “an” should generally be interpreted to include one or more described items. Accordingly, phrases such as “a device configured to” are intended to include one or more recited devices. Such one or more recited devices can also be collectively configured to carry out the stated recitations. For example, “a processor configured to carry out recitations A, B and C” can include a first processor configured to carry out recitation A working in conjunction with a second processor configured to carry out recitations B and C.

(76) While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, certain embodiments of the inventions described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others.

Claims

1. An automated first responder system, the system comprising: a patient monitor comprising a hardware processor configured to derive physiological parameter values from physiological signals obtained from one or more physiological sensors coupled with a patient; a gateway device, the patient monitor in communication with the gateway device over a network, wherein the gateway device is in communication with at least one clinician device and an electronic medical record (EMR) database storing patient data, the patient data comprising lab results of the patient, wherein the patient monitor is configured to access the patient data stored in the EMR database through the gateway device; and an autonomous delivery device in communication with the patient monitor, the autonomous delivery device comprising a substance; wherein the hardware processor of the patient monitor is configured to: analyze the physiological parameter values and the patient data based on a diagnostic model; determine that the physiological parameter values and the patient data correspond with the diagnostic model; in response to determining that it is a first time the physiological parameter values and the patient data correspond with the diagnostic model, initiate a

treatment workflow specific to the diagnostic model, the treatment workflow includes instructing the autonomous delivery device to automatically administer a first dose of the substance to the patient; in response to determining that it is not the first time the physiological parameter values and the patient data correspond with the diagnostic model, output an alert to the at least one clinician device so that a clinician can decide whether to administer a second dose of the substance to the patient, the second dose being greater than the first dose.

2. The system of claim 1, wherein the network is a hospital network.
3. The system of claim 1, wherein the network is a local area network, a wide area network, an intranet, the Internet, or any combinations thereof.
4. The system of claim 1, wherein the one or more physiological sensors comprise one or more of optical sensors, acoustic respiratory sensors, ECG sensors, brainwave measurement sensors, and/or blood pressure sensors.
5. The system of claim 1, wherein the at least one clinician device comprises one or more of a tablet, a pager, a smartphone, a laptop, a computer, and/or a nurses' station computers.
6. The system of claim 1, wherein the alert to the at least one clinician device is further configured to allow the clinician to perform an additional action associated with the treatment workflow.
7. The system of claim 1, wherein, in response to no response from the at least one clinician device within a predetermined time period, the hardware processor of the patient monitor is configured to instruct the autonomous delivery device to automatically administer the second dose of the substance to the patient.
8. The system of claim 7, wherein the hardware processor of the patient monitor is configured to instruct the autonomous delivery device to automatically administer the second dose of the substance to the patient further based on the patient data stored in the EMR database.
9. The system of claim 1, wherein the hardware processor of the patient monitor is configured to output the alert to a plurality of clinician devices, a number of clinician devices alerted for the diagnostic model depending on a severity of the diagnostic model and/or a severity of the physiological parameter values of the patient.
10. The system of claim 1, wherein the hardware processor of the patient monitor is configured to output different levels of the alert to the at least one clinician device, a level of the alert sent to the at least one clinician device varying depending on a seriousness of the diagnostic model.
11. The system of claim 1, wherein the hardware processor of the patient monitor is further configured to analyze the patient data according to static thresholds and dynamic thresholds.
12. The system of claim 11, wherein the static thresholds comprise predetermined thresholds that are the same for a class of patients and the dynamic thresholds comprise thresholds that are based on the patient's mean or baseline physiological parameter values.
13. The system of claim 1, wherein the hardware processor of the patient monitor is further configured to analyze the patient data by identifying a trend in the physiological parameter values and determining whether the trend corresponds with predetermined trends in the diagnostic model.
14. The system of claim 1, wherein the diagnostic model represents: an opioid overdose diagnostic model, a hypovolemia diagnostic model, or a sepsis diagnostic model.
15. The system of claim 14, wherein the opioid overdose diagnostic model corresponds to a decrease in respiratory rate, a decrease in heart rate, and a decrease in blood pressure.
16. The system of claim 15, wherein the substance comprises naloxone.
17. The system of claim 14, wherein the hypovolemia diagnostic model corresponds to a decrease in blood pressure, an increase in heart rate, and stability in hemoglobin level.
18. The system of claim 17, wherein the substance comprises saline.
19. The system of claim 14, wherein the sepsis diagnostic model corresponds to a decrease in blood pressure, an increase in white blood cell count, and an increase in temperature.
20. The system of claim 19, wherein initiating the treatment workflow further comprises one or more of: alerting a clinician, ordering a lab test, ordering a drug, and/or ordering a radiology test.

